

Andrew J. Graven

 [ajgraven](#) |  [ajgraven](#) |  [andrewgraven.com](#) |  [andrew@graven.com](#)

SUMMARY

Caltech Mathematics PhD with research experience in inverse problems, dynamical systems, and nonlinear PDEs, and a track record of translating mathematical ideas into high-performance computational tools. Developed GPU-accelerated software at NASA JPL for large-scale orbital analysis and mission design, including production software achieving up to $10^6 \times$ speedups.

EDUCATION

Caltech Mathematics PhD (2026)

Advisor: Nikolai G. Makarov

DOD National Defense Science Engineering Graduate (NDSEG) Research Fellow;

S. R. Johnson Dissertation Award; S. R. Johnson Graduate Excellence Prize

Cornell Mathematics BA (2021)

Summa Cum Laude, Distinction in all subjects

SKILLS & LEADERSHIP

Languages Python, C, CUDA/C, MATLAB, JavaScript, Mathematica.

Technical GPU & parallel computing, numerical optimization, numerical linear algebra, scientific computing

Tools Linux, Git, OpenMP, Agentic Coding

Leadership **Eagle Scout**; Caltech water polo goalie & coach - **character award** established in my name.

EXPERIENCE

Caltech – Doctoral Researcher, Mathematics

2021 - 2026

- Conducted original research in inverse problems, dynamical systems, and nonlinear PDEs, developing mathematical methods for recovering the geometry and evolution of domains from analytic data.
- Introduced new reconstruction techniques based on the Faber transform, conformal mapping, and potential theory; obtained classification and uniqueness results for quadrature domains and extended the framework to weighted domains connected with Hele-Shaw flow and Saffman-Taylor fingering.

NASA JPL – SURF Fellow

Summer 2018, 2019, 2020

2020 Delivered a production software tool (**CUDA/C**) for NASA to predict long-term satellite-ground station availability for satellite constellations. Achieved a $10^6 \times$ **speed-up** over classical algorithms. Presented [results](#) at the *AAS Space Flight Mechanics Conference (2021)*.

2019 Built and implemented a GPU-accelerated (CUDA/C) tool to scale interplanetary spacecraft tour design, accelerating evaluation/search of candidate tours in support of the *Europa Clipper* mission.

2018 Developed an ergodic theory-based approach to estimate long-term statistical behavior of satellite orbits, **enabling rapid forecasting without brute-force long-time simulation**. Presented [results](#) at the *Astrodynamics Specialist Conference (2019)*.

SELECTED RESEARCH

Analysis of Log-Weighted Quadrature Domains - 2026. Developed reconstruction and classification methods for generalized quadrature domains, with connections to nonlinear Hele-Shaw/Laplacian growth. ([pdf](#))

Quadrature Domains and the Faber Transform - 2025. Developed a Faber-transform framework for reconstructing quadrature domains from analytic data, yielding new classification and uniqueness results. ([pdf](#))

An Elementary Proof of Poincaré's Last Geometric Theorem - 2021. Developed a new elementary proof of the Poincaré-Birkhoff fixed point theorem using invariant curve classification and differential topology. ([pdf](#))

A Rapid Method for Orbital Coverage Statistics With J2 Using Ergodic Theory - 2021. Developed a method of predicting long-term satellite statistics without brute-force trajectory propagation (AAS). ([pdf](#))

Density of States Graph Kernels - 2021. Developed spectral methods for comparing graphs using density of states; coauthored peer-reviewed work (SIAM SDM) in numerical linear algebra and machine learning. ([pdf](#))